

Quantitative Comparisons

ANSWER KEY



Numeric comparisons

- 1 Column A: 7×8 . Column B: $6 \times 9 + 2$. Compare the two columns: which is greater, are they equal, or is it impossible to determine?

$$A = 56. B = 54 + 2 = 56. A = B.$$

- 2 Column A: $123 + 456$. Column B: $700 - 120$. Compare the two columns.

$$A = 579. B = 580. B > A.$$

- 3 Column A: 9^2 . Column B: $8^2 + 15$. Compare the two columns.

$$A = 81. B = 64 + 15 = 79. A > B.$$

- 4 Column A: $(-3)^2$. Column B: -3^2 . Compare the two columns.

$$A = (-3) \times (-3) = 9. B = -(3^2) = -9 \text{ (no parentheses around the base). } A > B.$$

Fraction and decimal comparisons

- 5 Column A: $\frac{3}{7}$. Column B: 0.43. Compare the two columns.

$$\frac{3}{7} \approx 0.4286, \text{ which is less than } 0.43. B > A.$$

- 6 Column A: $\frac{7}{12}$. Column B: 0.58. Compare the two columns.

$$\frac{7}{12} \approx 0.5833, \text{ which is greater than } 0.58. A > B.$$

- 7 Column A: $2\frac{3}{8}$. Column B: $\frac{19}{8}$. Compare the two columns.

$$2\frac{3}{8} = \frac{19}{8}. A = B.$$

- 8 Column A: 0.6. Column B: $\frac{5}{9}$. Compare the two columns.

$$\frac{5}{9} \approx 0.5556, \text{ which is less than } 0.6. A > B.$$

Percent comparisons

- 9 Column A: 25% of 160. Column B: 20% of 200. Compare the two columns.

$$A = 0.25 \times 160 = 40. B = 0.20 \times 200 = 40. A = B.$$

- 10 Column A: 30% of 90. Column B: 40% of 60. Compare the two columns.

$$A = 0.3 \times 90 = 27. B = 0.4 \times 60 = 24. A > B.$$

- 11 Column A: the result of increasing 100 by 10% twice in a row. Column B: the result of increasing 100 by 20% once. Compare the two columns.

$$A = 100 \times 1.1 \times 1.1 = 121. B = 100 \times 1.2 = 120. A > B.$$

- 12 Column A: 15% of 80. Column B: 12% of 100. Compare the two columns.

$$A = 0.15 \times 80 = 12. B = 0.12 \times 100 = 12. A = B.$$

Exponent and root comparisons

- 13 Column A: $\sqrt{50}$. Column B: 7.1. Compare the two columns.

$$\sqrt{50} \approx 7.07, \text{ which is less than } 7.1. B > A.$$

- 14 Column A: 2^{10} . Column B: 10^3 . Compare the two columns.

$$A = 1,024. B = 1,000. A > B.$$

- 15 Column A: $\sqrt{0.09}$. Column B: 0.3. Compare the two columns.

$$\sqrt{0.09} = 0.3. A = B.$$

- 16 Column A: 3^4 . Column B: 4^3 . Compare the two columns.

$$A = 81. B = 64. A > B.$$

Algebraic comparisons at a fixed value

- 17 Given $x = 5$. Column A: $2x + 3$. Column B: $x + 8$. Compare the two columns.

$$A = 2(5) + 3 = 13. B = 5 + 8 = 13. A = B.$$

- 18 Given $x = 4$. Column A: $(x + 2)^2$. Column B: $x^2 + 4x + 4$. Compare the two columns.

$$(x + 2)^2 \text{ expands to } x^2 + 4x + 4 \text{ for every value of } x, \text{ so the columns are always equal. } A = B.$$

- 19 Given $n = -2$. Column A: n^2 . Column B: n^3 . Compare the two columns.

$$A = (-2)^2 = 4. B = (-2)^3 = -8. A > B.$$

- 20 Given $x = 3$. Column A: $5x - 2$. Column B: $2(x + 5)$. Compare the two columns.

$$A = 5(3) - 2 = 13. B = 2(3 + 5) = 16. B > A.$$

Comparisons requiring testing values

- 21** For any real number x : Column A: x . Column B: x^2 . Compare the two columns.
If $x = 2$: $B > A$. If $x = \frac{1}{2}$: $A > B$. If $x = 1$: $A = B$. The relationship changes, so it cannot be determined.
- 22** For any real number x : Column A: $x + 5$. Column B: $2x$. Compare the two columns.
If $x = 5$: $A = B$. If $x = 0$: $A > B$. If $x = 10$: $B > A$. The relationship changes, so it cannot be determined.
- 23** For any real number x : Column A: $|x|$. Column B: x . Compare the two columns.
If $x \geq 0$: $A = B$. If $x < 0$ (e.g. $x = -3$): $A = 3$, $B = -3$, so $A > B$. Since the relationship is not the same in every case, it cannot be determined.
- 24** Given only that n is an integer: Column A: n^2 . Column B: 4. Compare the two columns.
If $n = 1$: $A = 1 < 4$. If $n = 3$: $A = 9 > 4$. The relationship changes, so it cannot be determined.
- 25** For any real number x : Column A: $(x - 3)^2$. Column B: $x^2 - 6x + 9$. Compare the two columns.
Expanding: $(x - 3)^2 = x^2 - 6x + 9$ for every value of x , so the columns are always equal. $A = B$.
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Geometry comparisons

- 26** Column A: the perimeter of a square with side 6 cm. Column B: the perimeter of a rectangle with length 8 cm and width 3 cm. Compare the two columns.
 $A = 4 \times 6 = 24$ cm. $B = 2(8 + 3) = 22$ cm. $A > B$.
- 27** Column A: the area of a circle with radius 4 cm (use $\pi \approx 3.14$). Column B: the area of a square with side 7 cm. Compare the two columns.
 $A = 3.14 \times 4^2 = 3.14 \times 16 = 50.24$ cm². $B = 7^2 = 49$ cm². $A > B$.
- 28** Column A: the sum of the interior angles of a pentagon. Column B: the sum of the interior angles of two triangles. Compare the two columns.
 $A = (5 - 2) \times 180^\circ = 540^\circ$. $B = 2 \times 180^\circ = 360^\circ$. $A > B$.
- 29** Column A: the hypotenuse of a right triangle with legs 5 and 12. Column B: 14. Compare the two columns.
 $A = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$. $B = 14$. $B > A$.
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Ratio and proportion comparisons

- 30** Column A: the value of x , if $x : 12 = 3 : 4$. Column B: 10. Compare the two columns.

$$x = \frac{3 \times 12}{4} = 9. B = 10. B > A.$$

- 31** A recipe uses flour and sugar in the ratio 3 : 1. Column A: the sugar needed for 240 g of flour. Column B: 90 g. Compare the two columns.

$$\text{Sugar} = 240 \div 3 = 80 \text{ g}. B = 90 \text{ g}. B > A.$$

- 32** Column A: the sum of the parts in the simplified form of the ratio 18 : 24. Column B: 8. Compare the two columns.

$$18 : 24 \text{ simplifies to } 3 : 4, \text{ so the sum of parts is } 3 + 4 = 7. B = 8. B > A.$$

- 33** Two numbers are in the ratio 5 : 3, and their sum is 96. Column A: the larger number. Column B: 61. Compare the two columns.

$$\text{Total parts} = 5 + 3 = 8, \text{ each part} = 12. \text{ Larger number} = 5 \times 12 = 60. B = 61. B > A.$$

Data and probability comparisons

- 34** A data set is 4, 7, 7, 9, 13. Column A: the mean. Column B: the median. Compare the two columns.

$$\text{Mean} = \frac{4 + 7 + 7 + 9 + 13}{5} = \frac{40}{5} = 8. \text{ Median (middle value)} = 7. A > B.$$

- 35** Column A: the range of the data set 3, 8, 15, 22. Column B: 20. Compare the two columns.

$$\text{Range} = 22 - 3 = 19. B = 20. B > A.$$

- 36** A bag contains 3 red balls and 5 blue balls. Column A: the probability of drawing a red ball. Column B: the probability of drawing a blue ball. Compare the two columns.

$$P(\text{red}) = \frac{3}{8}. P(\text{blue}) = \frac{5}{8}. B > A.$$

Word-problem comparisons

- 37** Column A: the total cost of 4 items at 15 riyals each, after a 10% discount on the total. Column B: 55 riyals. Compare the two columns.

$$\text{Total before discount} = 4 \times 15 = 60 \text{ riyals}. \text{ After discount: } 60 \times 0.9 = 54 \text{ riyals}. B = 55. B > A.$$

- 38** Ali's age is 3 years more than twice Sara's age. Sara is 8. Column A: Ali's age. Column B: 20. Compare the two columns.

$$\text{Ali} = 2(8) + 3 = 19. B = 20. B > A.$$

- 39 Column A: the simple interest earned on 2,000 riyals at 5% per year for 3 years. Column B: 350 riyals. Compare the two columns.

$$I = 2,000 \times 0.05 \times 3 = 300 \text{ riyals. } B = 350. B > A.$$

- 40 A car travels 240 km using 20 liters of fuel. Column A: the distance travelled per liter. Column B: 12 km per liter. Compare the two columns.

$$\text{Distance per liter} = 240 \div 20 = 12 \text{ km/liter. } B = 12. A = B.$$

Comparisons with inequalities and ranges

- 41 Given that $5 < x < 9$. Column A: x . Column B: 7. Compare the two columns.

If $x = 6$: $A < B$. If $x = 8$: $A > B$. The relationship changes, so it cannot be determined.

- 42 Given that $x > 10$. Column A: x . Column B: 10. Compare the two columns.

Since x is always greater than 10, $A > B$ in every case.

- 43 Given that $0 < x < 1$. Column A: x . Column B: x^2 . Compare the two columns.

For any value strictly between 0 and 1 (e.g. $x = 0.5$: $x^2 = 0.25 < 0.5$), squaring makes the value smaller. $A > B$ in every case.

- 44 Given that $-5 \leq x \leq -1$. Column A: x^2 . Column B: 10. Compare the two columns.

If $x = -1$: $A = 1 < 10$. If $x = -5$: $A = 25 > 10$. The relationship changes, so it cannot be determined.

More geometry comparisons

- 45 Column A: the volume of a cube with edge 4 cm. Column B: the volume of a rectangular box 3 cm $\times$ 4 cm $\times$ 6 cm. Compare the two columns.

$$A = 4^3 = 64 \text{ cm}^3. B = 3 \times 4 \times 6 = 72 \text{ cm}^3. B > A.$$

- 46 Column A: the circumference of a circle with radius 7 cm (use $\pi = \frac{22}{7}$). Column B: the perimeter of a square with side 11 cm. Compare the two columns.

$$A = 2 \times \frac{22}{7} \times 7 = 44 \text{ cm. } B = 4 \times 11 = 44 \text{ cm. } A = B.$$

- 47 Column A: the area of a triangle with base 10 and height 8. Column B: the area of a rectangle with length 9 and width 5. Compare the two columns.

$$A = \frac{1}{2} \times 10 \times 8 = 40. B = 9 \times 5 = 45. B > A.$$

- 48 Column A: the sum of the interior angles of a quadrilateral. Column B: three times the sum of the interior angles of a triangle. Compare the two columns.

$$A = 360^\circ. B = 3 \times 180^\circ = 540^\circ. B > A.$$

MCQ — Numeric comparisons

- 49 Column A: 6×7 . Column B: $5 \times 9 - 3$. Compare the two columns.

- a. The two columns are equal ✓
- b. Column B is greater
- c. The relationship cannot be determined from the given information
- d. Column A is greater

$$42 \text{ vs } 45 - 3 = 42: \text{equal.}$$

- 50 Column A: $345 + 678$. Column B: $900 + 100$. Compare the two columns.

- a. Column A is greater ✓
- b. The relationship cannot be determined from the given information
- c. Column B is greater
- d. The two columns are equal

$$1,023 \text{ vs } 1,000: \text{Column A is greater.}$$

- 51 Column A: 11^2 . Column B: $10^2 + 23$. Compare the two columns.

- a. Column B is greater ✓
- b. Column A is greater
- c. The relationship cannot be determined from the given information
- d. The two columns are equal

$$121 \text{ vs } 123: \text{Column B is greater.}$$

- 52 Column A: $(-4)^2$. Column B: -4^2 . Compare the two columns.
- a. The relationship cannot be determined from the given information
 - b. The two columns are equal
 - c. Column B is greater
 - d. **Column A is greater ✓**

16 vs -16 : Column A is greater.

MCQ — Fraction and decimal comparisons

- 53 Column A: $\frac{5}{8}$. Column B: 0.62. Compare the two columns.
- a. The relationship cannot be determined from the given information
 - b. Column B is greater
 - c. The two columns are equal
 - d. **Column A is greater ✓**

$\frac{5}{8} = 0.625$, which is greater than 0.62.

- 54 Column A: $\frac{9}{16}$. Column B: 0.5625. Compare the two columns.
- a. Column A is greater
 - b. Column B is greater
 - c. **The two columns are equal ✓**
 - d. The relationship cannot be determined from the given information

$\frac{9}{16} = 0.5625$ exactly: equal.

55 Column A: $3\frac{1}{6}$. Column B: $\frac{19}{6}$. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. Column B is greater
- c. Column A is greater
- d. The two columns are equal ✓

$$3\frac{1}{6} = \frac{19}{6}: \text{equal.}$$

56 Column A: 0.7. Column B: $\frac{5}{7}$. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. The two columns are equal
- c. Column B is greater ✓
- d. Column A is greater

$$\frac{5}{7} \approx 0.714, \text{ which is greater than } 0.7.$$

MCQ — Percent comparisons

57 Column A: 20% of 150. Column B: 25% of 120. Compare the two columns.

- a. The two columns are equal ✓
- b. Column A is greater
- c. Column B is greater
- d. The relationship cannot be determined from the given information

$$30 \text{ vs } 30: \text{equal.}$$

58 Column A: 45% of 80. Column B: 40% of 100. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. The two columns are equal
- c. Column A is greater
- d. **Column B is greater ✓**

36 vs 40: Column B is greater.

59 Column A: the result of decreasing 100 by 20%, then increasing that result by 20%. Column B: 96.

- a. The relationship cannot be determined from the given information
- b. **The two columns are equal ✓**
- c. Column A is greater
- d. Column B is greater

$100 \times 0.8 \times 1.2 = 96$: equal.

60 Column A: 18% of 50. Column B: 9% of 90. Compare the two columns.

- a. Column B is greater
- b. The relationship cannot be determined from the given information
- c. **Column A is greater ✓**
- d. The two columns are equal

9 vs 8.1: Column A is greater.

MCQ — Exponent and root comparisons

61 Column A: $\sqrt{72}$. Column B: 8.5. Compare the two columns.

- a. **Column B is greater ✓**
- b. Column A is greater
- c. The two columns are equal
- d. The relationship cannot be determined from the given information

$\sqrt{72} \approx 8.49$, which is less than 8.5.

62 Column A: 3^5 . Column B: 5^3 . Compare the two columns.

- a. **Column A is greater ✓**
- b. The two columns are equal
- c. Column B is greater
- d. The relationship cannot be determined from the given information

243 vs 125: Column A is greater.

63 Column A: $\sqrt{0.16}$. Column B: 0.4. Compare the two columns.

- a. Column A is greater
- b. Column B is greater
- c. The relationship cannot be determined from the given information
- d. **The two columns are equal ✓**

$\sqrt{0.16} = 0.4$ exactly: equal.

64 Column A: 2^6 . Column B: 6^2 . Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. **Column A is greater ✓**
- c. Column B is greater
- d. The two columns are equal

64 vs 36: Column A is greater.

MCQ — Algebraic comparisons at a fixed value

65 Given $x = 6$. Column A: $3x - 4$. Column B: $x + 10$. Compare the two columns.

- a. Column B is greater ✓
- b. Column A is greater
- c. The relationship cannot be determined from the given information
- d. The two columns are equal

14 vs 16: Column B is greater.

66 Given $x = 3$. Column A: $(x + 1)^2$. Column B: $x^2 + 2x + 1$. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. Column B is greater
- c. Column A is greater
- d. The two columns are equal ✓

Both equal 16 (this is an identity, true for every x): equal.

67 Given $n = -3$. Column A: n^2 . Column B: n^3 . Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. Column B is greater
- c. The two columns are equal
- d. Column A is greater ✓

9 vs -27 : Column A is greater.

68 Given $x = 2$. Column A: $4x + 1$. Column B: $3(x + 2)$. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. Column A is greater
- c. The two columns are equal
- d. **Column B is greater ✓**

9 vs 12: Column B is greater.

MCQ — Comparisons requiring testing values

69 For any real number x : Column A: x^2 . Column B: 0. Compare the two columns.

- a. **The relationship cannot be determined from the given information ✓**
- b. The two columns are equal
- c. Column A is greater
- d. Column B is greater

x^2 is always ≥ 0 , but could equal 0 (at $x = 0$) or exceed it — not always the same relationship.

70 For any integer n : Column A: n . Column B: $-n$. Compare the two columns.

- a. The two columns are equal
- b. **The relationship cannot be determined from the given information ✓**
- c. Column B is greater
- d. Column A is greater

If $n > 0$, A is greater; if $n < 0$, B is greater; if $n = 0$, they are equal — it varies.

71 For any real number x with $x \neq 0$: Column A: x^2 . Column B: 0. Compare the two columns.

- a. Column B is greater
- b. Column A is greater ✓**
- c. The two columns are equal
- d. The relationship cannot be determined from the given information

Since $x \neq 0$, x^2 is always strictly positive, so Column A is always greater.

72 For any real number x : Column A: $(x + 2)^2$. Column B: $x^2 + 4x + 4$. Compare the two columns.

- a. The two columns are equal ✓**
- b. Column A is greater
- c. The relationship cannot be determined from the given information
- d. Column B is greater

$(x + 2)^2 = x^2 + 4x + 4$ for every x : equal.

73 Given only that m is a positive integer: Column A: m . Column B: m^2 . Compare the two columns.

- a. Column B is greater
- b. Column A is greater
- c. The two columns are equal
- d. The relationship cannot be determined from the given information ✓**

At $m = 1$ they are equal; for $m > 1$, Column B is greater — it varies.

74 For any real number x : Column A: $-x$. Column B: $x - 1$. Compare the two columns.

- a. The two columns are equal
- b. Column A is greater
- c. The relationship cannot be determined from the given information ✓**
- d. Column B is greater

$-x - (x - 1) = 1 - 2x$, whose sign depends on x — it varies.

MCQ — Geometry comparisons

75 Column A: the circumference of the circle shown (use $\pi \approx 3.14$). Column B: 32.

- a. The relationship cannot be determined from the given information
- b. Column A is greater
- c. The two columns are equal
- d. **Column B is greater ✓**

$C = 2(3.14)(5) = 31.4$, which is less than 32.

76 Column A: the area of the rectangle shown. Column B: the perimeter of a square with side 7.

- a. The relationship cannot be determined from the given information
- b. The two columns are equal
- c. Column B is greater
- d. **Column A is greater ✓**

Rectangle area = $9 \times 4 = 36$; square perimeter = $4 \times 7 = 28$. Column A is greater.

77 Column A: the sum of the interior angles of a hexagon. Column B: the sum of the interior angles of two triangles.

- a. The two columns are equal
- b. The relationship cannot be determined from the given information
- c. **Column A is greater ✓**
- d. Column B is greater

Hexagon = 720° ; two triangles = $180 + 180 = 360^\circ$. Column A is greater.

78 Column A: the hypotenuse of the right triangle shown. Column B: 16.

- a. The relationship cannot be determined from the given information
- b. Column B is greater ✓**
- c. The two columns are equal
- d. Column A is greater

$$\sqrt{9^2 + 12^2} = \sqrt{225} = 15, \text{ which is less than } 16.$$

79 Column A: the volume of a cylinder with radius 3 and height 7 (use $\pi \approx 3.14$). Column B: 200.

- a. The relationship cannot be determined from the given information
- b. Column A is greater
- c. The two columns are equal
- d. Column B is greater ✓**

$$V = 3.14 \times 9 \times 7 = 197.82, \text{ which is less than } 200.$$

MCQ — Ratio and proportion comparisons

80 Column A: the value of x , if $x : 15 = 2 : 5$. Column B: 7.

- a. The relationship cannot be determined from the given information
- b. The two columns are equal
- c. Column A is greater
- d. Column B is greater ✓**

$$x = 15 \times \frac{2}{5} = 6, \text{ which is less than } 7.$$

81 Two numbers are in the ratio 4 : 7, and their sum is 99. Column A: the smaller number. Column B: 40.

- a. The two columns are equal
- b. The relationship cannot be determined from the given information
- c. Column A is greater
- d. Column B is greater ✓**

Each part = $99 \div 11 = 9$; smaller = $4 \times 9 = 36$, which is less than 40.

82 Column A: the sum of the parts in the simplified form of the ratio 24 : 36. Column B: 5.

- a. Column B is greater
- b. The relationship cannot be determined from the given information
- c. Column A is greater
- d. The two columns are equal ✓**

24 : 36 simplifies to 2 : 3; $2 + 3 = 5$: equal.

83 A recipe uses water and rice in the ratio 2 : 1. Column A: the water needed for 300 g of rice. Column B: 550 g.

- a. Column B is greater
- b. Column A is greater ✓**
- c. The two columns are equal
- d. The relationship cannot be determined from the given information

Water = $2 \times 300 = 600$ g, which is greater than 550 g.

MCQ — Data and probability comparisons

84 A data set is 5, 8, 8, 10, 14. Column A: the mean. Column B: the median.

- a. The two columns are equal
- b. Column B is greater
- c. **Column A is greater ✓**
- d. The relationship cannot be determined from the given information

Mean = $\frac{45}{5} = 9$; median = 8. Column A is greater.

85 Column A: the range of the data set 6, 11, 19, 25. Column B: 19.

- a. The relationship cannot be determined from the given information
- b. Column B is greater
- c. **The two columns are equal ✓**
- d. Column A is greater

Range = $25 - 6 = 19$: equal.

86 A bag contains 4 green balls and 9 yellow balls. Column A: the probability of drawing a green ball. Column B: 0.4.

- a. **Column B is greater ✓**
- b. The relationship cannot be determined from the given information
- c. The two columns are equal
- d. Column A is greater

$P = \frac{4}{13} \approx 0.31$, which is less than 0.4: Column B is greater.

87 A data set is 2, 4, 4, 4, 6. Column A: the mode. Column B: the mean.

- a. Column A is greater
- b. The relationship cannot be determined from the given information
- c. **The two columns are equal ✓**
- d. Column B is greater

Mode = 4; mean = $\frac{20}{5} = 4$: equal.

MCQ — Word-problem comparisons

88 Column A: the total cost of 5 items at 20 riyals each, after a 15% discount on the total. Column B: 90.

- a. The two columns are equal
- b. Column A is greater
- c. The relationship cannot be determined from the given information
- d. **Column B is greater ✓**

$5 \times 20 = 100$; after discount = $100 \times 0.85 = 85$, which is less than 90.

89 Layla's age is 4 years less than three times Huda's age. Huda is 9. Column A: Layla's age. Column B: 25.

- a. Column A is greater
- b. The two columns are equal
- c. **Column B is greater ✓**
- d. The relationship cannot be determined from the given information

Layla = $3(9) - 4 = 23$, which is less than 25.

90 Column A: the simple interest on 3,000 riyals at 4% per year for 2 years. Column B: 250.

- a. The two columns are equal
- b. The relationship cannot be determined from the given information
- c. Column A is greater
- d. **Column B is greater ✓**

$I = 3,000 \times 0.04 \times 2 = 240$, which is less than 250.

91 A car travels 300 km using 25 liters of fuel. Column A: the distance travelled per liter. Column B: 12 km.

- a. **The two columns are equal ✓**
- b. The relationship cannot be determined from the given information
- c. Column B is greater
- d. Column A is greater

$300 \div 25 = 12$ km per liter: equal.

MCQ — Comparisons with inequalities and ranges

92 Given that $3 < x < 8$. Column A: x . Column B: 5.

- a. **The relationship cannot be determined from the given information ✓**
- b. The two columns are equal
- c. Column A is greater
- d. Column B is greater

x could be less than, equal to, or greater than 5 within this range — it varies.

93 Given that $x < -4$. Column A: x . Column B: -4 .

- a. Column A is greater
- b. Column B is greater ✓**
- c. The two columns are equal
- d. The relationship cannot be determined from the given information

Every value less than -4 is smaller than -4 , so Column B is always greater.

94 Given that $-3 \leq x \leq 2$. Column A: x^2 . Column B: 9.

- a. Column B is greater
- b. Column A is greater
- c. The two columns are equal
- d. The relationship cannot be determined from the given information ✓**

At $x = -3$, $x^2 = 9$ (equal); at $x = 0$, $x^2 = 0$ (Column B greater) — it varies.

95 Given that $x \geq 7$. Column A: x^2 . Column B: 49.

- a. The relationship cannot be determined from the given information ✓**
- b. The two columns are equal
- c. Column B is greater
- d. Column A is greater

At $x = 7$ they are equal; for $x > 7$, Column A is greater — it varies.

MCQ — More geometry comparisons

96 Column A: the volume of the cube shown. Column B: the volume of a rectangular box $4 \times 5 \times 7$.

- a. The relationship cannot be determined from the given information
- b. Column A is greater
- c. **Column B is greater ✓**
- d. The two columns are equal

Cube $= 5^3 = 125$; box $= 4 \times 5 \times 7 = 140$. Column B is greater.

97 Column A: the circumference of a circle with radius 7 cm (use $\pi = \frac{22}{7}$). Column B: the area of a square with side 11 cm.

- a. The two columns are equal
- b. Column A is greater
- c. **Column B is greater ✓**
- d. The relationship cannot be determined from the given information

Circumference $= 2 \times \frac{22}{7} \times 7 = 44$; square area $= 121$. Column B is greater.

98 Column A: the sum of the interior angles of the octagon shown. Column B: four times the sum of the interior angles of a triangle.

- a. Column B is greater
- b. The two columns are equal
- c. **Column A is greater ✓**
- d. The relationship cannot be determined from the given information

Octagon $= (8 - 2) \times 180 = 1,080^\circ$; four triangles $= 4 \times 180 = 720^\circ$. Column A is greater.

Multiple choice — easy

99 Column A: 4×8 . Column B: $9 \times 5 - 8$. Compare the two columns.

- a. Column A is greater
- b. The relationship cannot be determined from the given information
- c. **Column B is greater ✓**
- d. The two columns are equal

32 vs $45 - 8 = 37$: Column B is greater.

100 Column A: 8×5 . Column B: $7 \times 8 - 15$. Compare the two columns.

- a. Column A is greater
- b. The two columns are equal
- c. The relationship cannot be determined from the given information
- d. **Column B is greater ✓**

40 vs $56 - 15 = 41$: Column B is greater.

101 Column A: 10×6 . Column B: $9 \times 11 - 3$. Compare the two columns.

- a. **Column B is greater ✓**
- b. The relationship cannot be determined from the given information
- c. Column A is greater
- d. The two columns are equal

60 vs $99 - 3 = 96$: Column B is greater.

102 Column A: 8×12 . Column B: $8 \times 3 + 1$. Compare the two columns.

- a. Column B is greater
- b. The two columns are equal
- c. **Column A is greater ✓**
- d. The relationship cannot be determined from the given information

96 vs $24 + 1 = 25$: Column A is greater.

103 Column A: 7×9 . Column B: $7 \times 3 - 11$. Compare the two columns.

- a. Column A is greater ✓**
- b. The two columns are equal
- c. The relationship cannot be determined from the given information
- d. Column B is greater

63 vs $21 - 11 = 10$: Column A is greater.

104 Column A: 4×11 . Column B: $10 \times 6 - 11$. Compare the two columns.

- a. The two columns are equal
- b. The relationship cannot be determined from the given information
- c. Column B is greater ✓**
- d. Column A is greater

44 vs $60 - 11 = 49$: Column B is greater.

105 Column A: 7×11 . Column B: $11 \times 8 + 9$. Compare the two columns.

- a. The two columns are equal
- b. The relationship cannot be determined from the given information
- c. Column A is greater
- d. Column B is greater ✓**

77 vs $88 + 9 = 97$: Column B is greater.

106 Column A: 9×7 . Column B: $9 \times 3 - 5$. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. Column B is greater
- c. **Column A is greater ✓**
- d. The two columns are equal

63 vs $27 - 5 = 22$: Column A is greater.

107 Column A: 6×11 . Column B: $11 \times 9 + 2$. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. Column A is greater
- c. The two columns are equal
- d. **Column B is greater ✓**

66 vs $99 + 2 = 101$: Column B is greater.

108 Column A: 8×4 . Column B: $3 \times 8 + 1$. Compare the two columns.

- a. **Column A is greater ✓**
- b. The two columns are equal
- c. Column B is greater
- d. The relationship cannot be determined from the given information

32 vs $24 + 1 = 25$: Column A is greater.

109 Column A: 7×6 . Column B: $4 \times 3 + 8$. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. **Column A is greater ✓**
- c. The two columns are equal
- d. Column B is greater

42 vs $12 + 8 = 20$: Column A is greater.

110 Column A: 11×10 . Column B: $6 \times 5 + 15$. Compare the two columns.

- a. The two columns are equal
- b. The relationship cannot be determined from the given information
- c. Column B is greater
- d. **Column A is greater ✓**

110 vs $30 + 15 = 45$: Column A is greater.

111 Column A: 12×12 . Column B: $9 \times 12 - 10$. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. The two columns are equal
- c. **Column A is greater ✓**
- d. Column B is greater

144 vs $108 - 10 = 98$: Column A is greater.

112 Column A: 8×3 . Column B: $4 \times 9 + 2$. Compare the two columns.

- a. Column A is greater
- b. **Column B is greater ✓**
- c. The two columns are equal
- d. The relationship cannot be determined from the given information

24 vs $36 + 2 = 38$: Column B is greater.

113 Column A: 3×11 . Column B: $12 \times 7 - 10$. Compare the two columns.

- a. The two columns are equal
- b. The relationship cannot be determined from the given information
- c. Column B is greater ✓**
- d. Column A is greater

33 vs $84 - 10 = 74$: Column B is greater.

Multiple choice — medium

114 Column A: 10% of 160. Column B: 20% of 40. Compare the two columns.

- a. The two columns are equal
- b. Column A is greater ✓**
- c. The relationship cannot be determined from the given information
- d. Column B is greater

$A = 0.1 \times 160 = 16$. $B = 0.2 \times 40 = 8$. Column A is greater.

115 Column A: 30% of 250. Column B: 80% of 60. Compare the two columns.

- a. Column B is greater
- b. The relationship cannot be determined from the given information
- c. Column A is greater ✓**
- d. The two columns are equal

$A = 0.3 \times 250 = 75$. $B = 0.8 \times 60 = 48$. Column A is greater.

116 Column A: 50% of 70. Column B: 60% of 250. Compare the two columns.

- a. Column A is greater
- b. The relationship cannot be determined from the given information
- c. Column B is greater ✓**
- d. The two columns are equal

$A = 0.5 \times 70 = 35$. $B = 0.6 \times 250 = 150$. Column B is greater.

117 Column A: 80% of 60. Column B: 30% of 120. Compare the two columns.

- a. Column B is greater
- b. The two columns are equal
- c. The relationship cannot be determined from the given information
- d. Column A is greater ✓**

$A = 0.8 \times 60 = 48$. $B = 0.3 \times 120 = 36$. Column A is greater.

118 Column A: 50% of 170. Column B: 30% of 250. Compare the two columns.

- a. Column A is greater ✓**
- b. The two columns are equal
- c. Column B is greater
- d. The relationship cannot be determined from the given information

$A = 0.5 \times 170 = 85$. $B = 0.3 \times 250 = 75$. Column A is greater.

119 Column A: 15% of 80. Column B: 20% of 290. Compare the two columns.

- a. Column A is greater
- b. The two columns are equal
- c. Column B is greater ✓**
- d. The relationship cannot be determined from the given information

$A = 0.15 \times 80 = 12$. $B = 0.2 \times 290 = 58$. Column B is greater.

120 Column A: 60% of 20. Column B: 20% of 60. Compare the two columns.

- a. Column B is greater
- b. The relationship cannot be determined from the given information
- c. Column A is greater
- d. The two columns are equal ✓**

$A = 0.6 \times 20 = 12$. $B = 0.2 \times 60 = 12$. The two columns are equal.

121 Column A: 75% of 60. Column B: 10% of 160. Compare the two columns.

- a. Column A is greater ✓**
- b. The relationship cannot be determined from the given information
- c. The two columns are equal
- d. Column B is greater

$A = 0.75 \times 60 = 45$. $B = 0.1 \times 160 = 16$. Column A is greater.

122 Column A: 60% of 150. Column B: 20% of 150. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. The two columns are equal
- c. Column A is greater ✓**
- d. Column B is greater

$A = 0.6 \times 150 = 90$. $B = 0.2 \times 150 = 30$. Column A is greater.

123 Column A: 50% of 190. Column B: 30% of 70. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. Column A is greater ✓**
- c. The two columns are equal
- d. Column B is greater

$A = 0.5 \times 190 = 95$. $B = 0.3 \times 70 = 21$. Column A is greater.

124 Column A: 75% of 200. Column B: 75% of 20. Compare the two columns.

- a. The two columns are equal
- b. The relationship cannot be determined from the given information
- c. Column A is greater ✓**
- d. Column B is greater

$A = 0.75 \times 200 = 150$. $B = 0.75 \times 20 = 15$. Column A is greater.

125 Column A: 10% of 130. Column B: 60% of 230. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. Column A is greater
- c. The two columns are equal
- d. Column B is greater ✓**

$A = 0.1 \times 130 = 13$. $B = 0.6 \times 230 = 138$. Column B is greater.

126 Column A: 30% of 280. Column B: 15% of 100. Compare the two columns.

- a. Column A is greater ✓**
- b. The relationship cannot be determined from the given information
- c. The two columns are equal
- d. Column B is greater

$A = 0.3 \times 280 = 84$. $B = 0.15 \times 100 = 15$. Column A is greater.

127 Column A: 20% of 140. Column B: 50% of 200. Compare the two columns.

- a. The two columns are equal
- b. The relationship cannot be determined from the given information
- c. Column B is greater ✓**
- d. Column A is greater

$A = 0.2 \times 140 = 28$. $B = 0.5 \times 200 = 100$. Column B is greater.

128 Column A: 30% of 210. Column B: 30% of 270. Compare the two columns.

- a. Column A is greater
- b. The two columns are equal
- c. Column B is greater ✓**
- d. The relationship cannot be determined from the given information

$A = 0.3 \times 210 = 63$. $B = 0.3 \times 270 = 81$. Column B is greater.

Multiple choice — hard

129 Given $x = 3$. Column A: $x + 4$. Column B: $2x + 5$. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. Column B is greater ✓**
- c. Column A is greater
- d. The two columns are equal

$A = (3) + 4 = 7$. $B = 2(3) + 5 = 11$. Column B is greater.

130 Given $x = -5$. Column A: x . Column B: $x - 9$. Compare the two columns.

- a. Column B is greater
- b. The relationship cannot be determined from the given information
- c. **Column A is greater ✓**
- d. The two columns are equal

$A = (-5) = -5$. $B = (-5) - 9 = -14$. Column A is greater.

131 Given $x = -9$. Column A: $2x - 2$. Column B: $5x - 2$. Compare the two columns.

- a. Column B is greater
- b. **Column A is greater ✓**
- c. The two columns are equal
- d. The relationship cannot be determined from the given information

$A = 2(-9) - 2 = -20$. $B = 5(-9) - 2 = -47$. Column A is greater.

132 Given $x = -3$. Column A: $3x + 9$. Column B: $4x + 9$. Compare the two columns.

- a. The two columns are equal
- b. Column B is greater
- c. **Column A is greater ✓**
- d. The relationship cannot be determined from the given information

$A = 3(-3) + 9 = 0$. $B = 4(-3) + 9 = -3$. Column A is greater.

133 Given $x = 8$. Column A: $2x + 5$. Column B: $2x + 9$. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. Column A is greater
- c. **Column B is greater ✓**
- d. The two columns are equal

$A = 2(8) + 5 = 21$. $B = 2(8) + 9 = 25$. Column B is greater.

134 For any real x . Column A: x^2 . Column B: 4. Compare the two columns.

- a. Column B is greater
- b. Column A is greater
- c. The relationship cannot be determined from the given information ✓**
- d. The two columns are equal

x^2 can be less than, equal to, or greater than 4 depending on x (e.g. $x = 0$ gives $x^2 = 0$; large x gives $x^2 > 4$), so the relationship is not constant.

135 For any nonzero integer n . Column A: n . Column B: $-n$. Compare the two columns.

- a. Column B is greater
- b. The two columns are equal
- c. Column A is greater
- d. The relationship cannot be determined from the given information ✓**

If $n > 0$, Column A is greater; if $n < 0$, Column B is greater — the relationship changes.

136 Given $x < 15$. Column A: x . Column B: 5. Compare the two columns.

- a. The relationship cannot be determined from the given information ✓**
- b. Column B is greater
- c. Column A is greater
- d. The two columns are equal

x could be less than 5 (e.g. $x = 0$) or greater than 5 (e.g. $x = 6$, still less than 15), so the relationship is not constant.

137 For any real m . Column A: $|m|$. Column B: m . Compare the two columns.

- a. Column B is greater
- b. The two columns are equal
- c. The relationship cannot be determined from the given information ✓**
- d. Column A is greater

If $m \geq 0$, the columns are equal; if $m < 0$, Column A is greater (since $|m| > 0 > m$) — the relationship is not constant.

138 For any real t . Column A: t^2 . Column B: t . Compare the two columns.

- a. The two columns are equal
- b. Column B is greater
- c. The relationship cannot be determined from the given information ✓**
- d. Column A is greater

If $t = 2$, $A = 4 > B$; if $t = 0.5$, $A = 0.25 < B$; if $t = 1$, $A = B$ — the relationship is not constant.

139 Column A: the area of a rectangle with length 13 cm and width 7 cm. Column B: the area of a square with side 11 cm.

- a. The two columns are equal
- b. Column A is greater
- c. Column B is greater ✓**
- d. The relationship cannot be determined from the given information

$A = 13 \times 7 = 91 \text{ cm}^2$. $B = 11^2 = 121 \text{ cm}^2$. Column B is greater.

140 Column A: the perimeter of a rectangle with length 5 cm and width 3 cm. Column B: the perimeter of a square with side 5 cm.

- a. Column B is greater ✓**
- b. The two columns are equal
- c. The relationship cannot be determined from the given information
- d. Column A is greater

$A = 2(5 + 3) = 16$ cm. $B = 4 \times 5 = 20$ cm. Column B is greater.

141 Column A: the perimeter of a rectangle with length 8 cm and width 5 cm. Column B: the perimeter of a square with side 14 cm.

- a. The two columns are equal
- b. Column B is greater ✓**
- c. The relationship cannot be determined from the given information
- d. Column A is greater

$A = 2(8 + 5) = 26$ cm. $B = 4 \times 14 = 56$ cm. Column B is greater.

142 Column A: the perimeter of a rectangle with length 4 cm and width 3 cm. Column B: the perimeter of a square with side 14 cm.

- a. Column B is greater ✓**
- b. The two columns are equal
- c. Column A is greater
- d. The relationship cannot be determined from the given information

$A = 2(4 + 3) = 14$ cm. $B = 4 \times 14 = 56$ cm. Column B is greater.

143 Column A: the area of a rectangle with length 6 cm and width 4 cm. Column B: the area of a square with side 4 cm.

- a. The two columns are equal
- b. The relationship cannot be determined from the given information
- c. **Column A is greater ✓**
- d. Column B is greater

$A = 6 \times 4 = 24 \text{ cm}^2$. $B = 4^2 = 16 \text{ cm}^2$. Column A is greater.